

1. Vision & Concept Transforming a Snapdragon-powered PC into an offline "AI Brain" for drones, completely offloading heavy computer vision tasks from the drone to the PC's NPU for zero-latency, cloud-free edge computing.

2. The Core Problem Small autonomous drones face strict payload and battery limits, making onboard AI processing nearly impossible. On the other hand, streaming live video to cloud-based AI introduces dangerous latency and fails entirely in remote field operations where internet connectivity is unavailable.

3. Our Edge-Compute Solution Instead of forcing the drone to process AI, it simply transmits a lightweight live camera feed to a ground-based Snapdragon PC via telemetry radio. The PC processes this video locally and instantly transmits lightweight flight commands (e.g., "HOVER" or "TURN") back to the drone's hardware.

4. Technical Architecture

- **AI Model:** We utilize a pre-quantized YOLO object detection model downloaded directly from the **Qualcomm AI Hub**.
- **Hardware Acceleration:** The model runs natively on the **Snapdragon Hexagon NPU** via ONNX Runtime, delivering ultra-fast inference while keeping the CPU/GPU free.
- **The Bridge:** A Python serial script translates the NPU's AI bounding-box outputs into physical hardware signals sent via a telemetry radio module (e.g., nRF24L01).

5. Real-World Impact

- **Battery Efficiency:** The drone remains incredibly lightweight, maximizing flight time.
- **Zero Latency & Privacy:** 100% offline processing guarantees instant reaction times and total data security.
- **True Hardware Integration:** Showcases Windows on ARM as a robust, real-world control hub for robotics.